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OSCILLATING POSITIVE EXPIRATORY PRESSURE DEVICE

Abstract

A respiratory treatment device comprising at least one chamber, a chamber inlet configured to receive exhaled air into the at least one chamber, at least one chamber outlet configured to permit exhaled air to exit the at least one chamber, and an exhalation flow path defined between the chamber inlet and the at least one chamber outlet. A restrictor member positioned in the exhalation flow path is moveable between a closed position, where a flow of exhaled air along the exhalation flow path is restricted, and an open position, where the flow of exhaled air along the exhalation flow path is less restricted. A vane in fluid communication with the exhalation flow path is operatively connected to the restrictor member and is configured to reciprocate between a first position and a second position in response to the flow of exhaled air along the exhalation flow path.

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Background/Summary

RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/350,097, filed on Jul. 11, 2023, pending, which is a continuation of U.S. application Ser. No. 17/324,448, filed on May 19, 2021, now U.S. Pat. No. 11,738,167, which is a continuation of U.S. application Ser. No. 15/903,761, filed on Feb. 23, 2018, now U.S. Pat. No. 11,040,167, which is a continuation of U.S. application Ser. No. 15/143,092, filed on Apr. 29, 2016, now U.S. Pat. No. 9,981,106, which is a continuation of U.S. application Ser. No. 13/489,894, filed on Jun. 6, 2012, now U.S. Pat. No. 9,358,417, which claims the benefit of U.S. Provisional Application No. 61/532,951, filed on Sep. 9, 2011, expired, and U.S. Provisional Application No. 61/493,816, filed on Jun. 6, 2011, expired, all of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a respiratory treatment device, and in particular, to an oscillating positive expiratory pressure (“OPEP”) device.

BACKGROUND

[0003] Each day, humans may produce upwards of 30 milliliters of sputum, which is a type of bronchial secretion. Normally, an effective cough is sufficient to loosen secretions and clear them from the body's airways. However, for individuals suffering from more significant bronchial obstructions, such as collapsed airways, a single cough may be insufficient to clear the obstructions.

[0004] OPEP therapy represents an effective bronchial hygiene technique for the removal of bronchial secretions in the human body and is an important aspect in the treatment and continuing care of patients with bronchial obstructions, such as those suffering from chronic obstructive lung disease. It is believed that OPEP therapy, or the oscillation of exhalation pressure at the mouth during exhalation, effectively transmits an oscillating back pressure to the lungs, thereby splitting open obstructed airways and loosening the secretions contributing to bronchial obstructions.

[0005] OPEP therapy is an attractive form of treatment because it can be easily taught to most patients, and such patients can assume responsibility for the administration of OPEP therapy throughout a hospitalization and also from home. To that end, a number of portable OPEP devices have been developed.

BRIEF SUMMARY

[0006] In order to provide an effective means for delivering OPEP therapy, a method and device for administering OPEP therapy is disclosed. In a first aspect, a respiratory treatment device includes a housing enclosing at least one chamber, a chamber inlet configured to receive exhaled air into the at least one chamber, at least one chamber outlet configured to permit exhaled air to exit the at least one chamber, and an exhalation flow path defined between the chamber inlet and the at least one chamber outlet. A restrictor member positioned in the exhalation flow path is moveable between a closed position, where a flow of exhaled air along the exhalation flow path is restricted, and an open position, where the flow of exhaled air along the exhalation flow path is less restricted. Furthermore, a vane in fluid communication with the exhalation flow path is operatively connected to the restrictor member and is configured to reciprocate between a first position and a second position in response to a flow of exhaled air along the exhalation flow path. The restrictor member and the vane are axially offset along a common axis of rotation.

[0007] In another aspect, in the first position, the vane is positioned to direct the flow of exhaled air to exit the at least one chamber through a first chamber outlet of the at least one chamber outlet, and in the second position, the vane is positioned to direct the flow of exhaled air to exit the at least one chamber through a second chamber outlet of the at least one chamber outlet.

[0008] In another aspect, the restrictor member is positioned in a first chamber, while the vane is positioned in a second chamber. The first chamber and the second chamber may be connected by an orifice, and the size of the orifice may be configured to change in response to the flow of exhaled air through the orifice. The restrictor member may be implemented as a butterfly valve. Also, the restrictor member may be operatively connected to the restrictor member by a shaft.

[0009] In yet another aspect, a face of the restrictor member is rotatable about an axis of rotation. The face of the restrictor member may also be radially offset from the axis of rotation. In addition, the face of the restrictor member may have a greater surface area positioned on one side of the shaft than the other side of the shaft.

[0010] In another aspect, an orientation of the chamber inlet is selectively adjustable.

[0011] In a further aspect, the respiratory treatment device includes a chamber inlet bypass configured to permit exhaled air into the at least one chamber without passing through the chamber inlet.

[0012] In another aspect, the respiratory treatment device includes a control port configured to permit exhaled air to exit the respiratory treatment device prior to entering the at least one chamber. The respiratory treatment device may also include a control port configured to permit exhaled air to exit the first chamber.

[0013] In a further aspect, the respiratory treatment device includes an inhalation port in fluid communication with a user interface. The respiratory treatment device may also include a one-way valve configured to permit air to flow through the inhalation port to the user interface upon inhalation. The inhalation port may be adapted to receive an aerosol medicament from an aerosol delivery device. The aerosol delivery device may be connected to the inhalation port.

[0014] In another aspect, the exhalation flow path is folded upon itself.

[0015] In another aspect, a respiratory treatment device includes a housing enclosing at least one chamber, a chamber inlet configured to receive exhaled air into the at least one chamber, at least one chamber outlet configured to permit exhaled air to exit the at least one chamber, and an exhalation flow path defined between the chamber inlet and the at least one chamber outlet. A restrictor member positioned in the exhalation flow path is moveable between a closed position, where a flow of exhaled air through the chamber inlet is restricted, and an open position, where the flow of exhaled air through the chamber inlet is less restricted. In addition, an orifice is positioned along the exhalation flow path through which exhaled air passes. A vane positioned adjacent the orifice is operatively connected to the restrictor member and is configured to reciprocate between a first position and a second position in response to an increased pressure adjacent the vane. The restrictor member moves between the closed position and the open position in response to the vane

reciprocating between the first position and the second position.

[0016] In a further aspect, the restrictor member may be positioned in a first chamber and the vane may be positioned in a second chamber, with the orifice separating the first and the second chamber. In addition, the size of the orifice may be configured to change in response to the flow of exhaled air through the orifice.

[0017] In another aspect, a respiratory treatment device includes a housing enclosing at least one chamber, a chamber inlet configured to receive exhaled air into the at least one chamber, at least one chamber outlet configured to permit exhaled air to exit the at least one chamber, and an exhalation flow path defined between the chamber inlet and the at least one chamber outlet. A restrictor member positioned in the exhalation flow path is moveable in response to a flow of exhaled air along the exhalation flow path between a closed position, where the flow of exhaled air along the exhalation flow path is restricted, and an open position, where the flow of exhaled air along the exhalation flow path is less restricted. Furthermore, a variable nozzle is positioned in the exhalation flow path such that the exhalation flow path passes through an orifice of the variable nozzle. A size of the orifice is configured to increase in response to the flow of exhaled air through the orifice.

[0018] In yet another aspect, the respiratory treatment device may comprise a vane positioned adjacent the orifice. The vane may be operatively connected to the restrictor member such that it is configured to move the restrictor member between the closed position and the open position in response to an increased pressure adjacent the vane.

[0019] In another aspect, the variable nozzle is positioned downstream from the restrictor member in the exhalation flow path.

[0020] In a further aspect, the orifice of the variable nozzle is substantially rectangular. The orifice of the variable nozzle may remain substantially rectangular after an increase in the size of the orifice in response to the flow of exhaled air through the orifice.

[0021] In yet another aspect, a method of performing OPEP therapy includes receiving a flow of exhaled air along an exhalation flow path defined between an inlet and an outlet of a respiratory treatment device, directing the flow of exhaled air toward a vane, and reciprocating the vane between a first position and a second position in response to the flow of exhaled air. The method further includes moving a restrictor member in response to the reciprocal movement of the vane between a closed position, where a flow of exhaled air through the chamber inlet is restricted, and an open position, where the flow of exhaled air is less restricted.

[0022] In another aspect, a method of performing OPEP therapy includes receiving a flow of exhaled air along an exhalation flow path defined between an inlet and an outlet of a respiratory treatment device, accelerating the flow of exhaled air through an orifice positioned along the exhalation flow path, and reciprocating a vane adjacent the orifice between a first position and a second position in response to the flow of exhaled air through the orifice. The method further includes moving a restrictor member in response to the reciprocal movement of the vane between a closed position, where a flow of exhaled air along the exhalation flow path is restricted, and an open position, where the flow of exhaled air along the exhalation flow path is less restricted. The method may also include changing a size of the orifice in response to the flow of exhaled air through the orifice.

[0023] In yet another aspect, a respiratory treatment device includes a housing enclosing a plurality of chambers, a first opening in the housing configured to transmit air exhaled into and air inhaled from the housing, a second opening in the housing configured to permit air exhaled into the first opening to exit the housing, and a third opening in the housing configured to permit air outside the housing to enter the housing upon inhalation at the first opening. An exhalation flow path is defined between the first opening and the second opening, and an inhalation flow path defined between the third opening and the first opening. A restrictor member is positioned in the exhalation flow path and the inhalation flow path, and is movable between a closed position, where a flow of air along

the exhalation flow path or the inhalation flow path is restricted, and an open position, where the flow of exhaled air along the exhalation flow path or the inhalation flow path is less restricted. A vane is in fluid communication with the exhalation flow path and the inhalation flow path. The vane is operatively connected to the restrictor member and configured to repeatedly reciprocate between a first position and a second position in response to the flow of air along the exhalation flow path or the inhalation flow path.

[0024] In a further aspect, the exhalation flow path and the inhalation flow path form an overlapping portion. The flow of air along the exhalation flow path and the inhalation flow path along the overlapping portion may be in the same direction. Furthermore, the restrictor member may be positioned in the overlapping portion, and the vane may be in fluid communication with the overlapping portion.

[0025] In another aspect, the restrictor member is positioned in a first chamber of the plurality of chambers, and the vane is positioned in a second chamber of the plurality of chambers. The flow of air through an inlet to the first chamber may be restricted when the restrictor member is in the closed position, and the flow of air through the inlet may be less restricted when the restrictor member is in the open position. In addition, the first chamber and the second chamber may be connected by an orifice. Furthermore, the vane may be positioned adjacent the orifice such that the vane is configured to move the restrictor member between the closed position and the open position in response to an increased pressure adjacent the vane.

[0026] In yet another aspect, the second opening includes a one-way exhalation valve configured to permit air exhaled into the housing to exit the housing upon exhalation at the first opening.

[0027] In another aspect, the third opening includes a one-way inhalation valve configured to permit air outside the housing to enter the housing upon inhalation at the first opening.

[0028] In an additional aspect, a one-way valve is positioned along the exhalation flow path between the first opening and the second opening, such that the one-way valve is configured to open in response to air exhaled into the first opening, and close in response to air inhaled through the first opening.

[0029] In a further aspect, a one-way valve is positioned along the inhalation flow path between the third opening and the first opening, such that the one-way valve is configured to open in response to air inhaled through the first opening, and close in response to air exhaled into the first opening.

[0030] In yet another aspect, the respiratory treatment device may include an inhalation port in fluid communication with a user interface, wherein the inhalation port is adapted to receive an medicament suitable for inhalation from an aerosol delivery device. The aerosol delivery device may be connected to the inhalation port.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0031] FIG. 1 is a front perspective view of an OPEP device;

[0032] FIG. 2 is a rear perspective view of the OPEP device of FIG. 1;

[0033] FIG. 3 is a cross-sectional perspective view taken along line III in FIG. 1 of the OPEP device shown without the internal components of the OPEP device;

[0034] FIG. 4 is an exploded view of the OPEP device of FIG. 1, shown with the internal components of the OPEP device;

[0035] FIG. 5 is a cross-sectional perspective view taken along line III in FIG. 1 of the OPEP device shown with the internal components of the OPEP device;

[0036] FIG. 6 is a different cross-sectional perspective view taken along line VI in FIG. 1 of the OPEP device shown with the internal components of the OPEP device;

[0037] FIG. 7 is a different cross-sectional perspective view taken along line VII in FIG. 1 of the

OPEP device shown with the internal components of the OPEP device;

[0038] FIG. **8** is a front perspective view of a restrictor member operatively connected to a vane;

[0039] FIG. **9** is a rear perspective view of the restrictor member operatively connected to the vane shown in FIG. **8**;

[0040] FIG. **10** is a front view of the restrictor member operatively connected to the vane shown in FIG. **8**;

[0041] FIG. **11** is a top view of the restrictor member operatively connected to the vane shown in FIG. **8**;

[0042] FIG. **12** is a front perspective view of a variable nozzle shown without the flow of exhaled air therethrough;

[0043] FIG. **13** is a rear perspective view of the variable nozzle of FIG. **12** shown without the flow of exhaled air therethrough;

[0044] FIG. **14** is a front perspective view of the variable nozzle of FIG. **12** shown with a high flow of exhaled air therethrough;

[0045] FIGS. **15A-C** are top phantom views of the OPEP device of FIG. **1** showing an exemplary illustration of the operation of the OPEP device of FIG. **1**;

[0046] FIG. **16** is a front perspective view of a different embodiment of a variable nozzle shown without the flow of exhaled air therethrough;

[0047] FIG. **17** is a rear perspective view of the variable nozzle of FIG. **16** shown without the flow of exhaled air therethrough;

[0048] FIG. **18** is a front perspective view of a second embodiment of an OPEP device;

[0049] FIG. **19** is a rear perspective view of the OPEP device of FIG. **18**;

[0050] FIG. **20** is an exploded view of the OPEP device of FIG. **18**, shown with the internal components of the OPEP device;

[0051] FIG. **21** is a cross-sectional view taken along line I in FIG. **18** of the OPEP device, shown with the internal components of the OPEP device;

[0052] FIG. **22** is a cross-sectional view taken along line II in FIG. **18** of the OPEP device, shown with the internal components of the OPEP device;

[0053] FIG. **23** is a cross-sectional view taken along line III in FIG. **18** of the OPEP device, shown with the internal components of the OPEP device;

[0054] FIG. **24** is a front perspective view of an adjustment mechanism of the OPEP device of FIG. **18**;

[0055] FIG. **25** is a rear perspective view of the adjustment mechanism of FIG. **24**;

[0056] FIG. **26** is a front perspective view of a restrictor member operatively connected to a vane for use in the OPEP device of FIG. **18**;

[0057] FIG. **27** is a front perspective view of the adjustment mechanism of FIG. **24** assembled with the restrictor member and the vane of FIG. **26**;

[0058] FIG. **28** is a partial cross-sectional view of the assembly of FIG. **27** within the OPEP device of FIG. **18**;

[0059] FIGS. **29A-B** are partial cross-sectional views illustrating installation of the assembly of FIG. **27** within the OPEP device of FIG. **18**;

[0060] FIG. **30** is a front view of the OPEP device of FIG. **18** illustrating an aspect of the adjustability of the OPEP device;

[0061] FIG. **31** is a partial cross-sectional view of the assembly of FIG. **27** within the OPEP device of FIG. **18**;

[0062] FIGS. **32A-B** are partial cross-sectional views taken along line III in FIG. **18** of the OPEP device, illustrating possible configurations of the OPEP device;

[0063] FIGS. **33A-B** are top phantom views illustrating the adjustability of the OPEP device of FIG. **18**;

[0064] FIGS. **34A-B** are top phantom views of the OPEP device of FIG. **18**, illustrating the

adjustability of the OPEP device;

[0065] FIG. **35** is a front perspective view of a third embodiment of an OPEP device;

[0066] FIG. **36** is a cross-sectional view taken along line I in FIG. **35** of the OPEP device;

[0067] FIG. **37** is a front perspective view of an adjustment mechanism of the OPEP device of FIG. **35** assembled with a restrictor member and a vane;

[0068] FIG. **38** is a rear perspective view of the assembly of FIG. **37**;

[0069] FIG. **39** is a front perspective view of a fourth embodiment of an OPEP device;

[0070] FIG. **40** is a cross-sectional view taken along line I in FIG. **39** of the OPEP device;

[0071] FIGS. **41A-B** are top phantom views of the OPEP device of FIG. **39**, illustrating the adjustability of the OPEP device;

[0072] FIG. **42** is a front perspective view of an alternative embodiment of the OPEP device of FIG. **1**;

[0073] FIG. **43** is a cross-sectional view taken along line I in FIG. **42** of the OPEP device;

[0074] FIG. **44** is a front perspective view of another alternative embodiment of the OPEP device of FIG. **1**;

[0075] FIG. **45** is a cross-sectional view taken along line I in FIG. **44** of the OPEP device;

[0076] FIG. **46** is a front perspective view of yet another alternative embodiment of the OPEP device of FIG. **1**;

[0077] FIG. **47** is a cross-sectional view taken along line I in FIG. **46** of the OPEP device;

[0078] FIG. **48** is a front perspective view of another embodiment of an OPEP device;

[0079] FIG. **49** is a rear perspective view of the OPEP device of FIG. **48**;

[0080] FIG. **50** is a perspective view of the bottom of the OPEP device of FIG. **48**;

[0081] FIG. **51** is an exploded view of the OPEP device of FIG. **48**;

[0082] FIG. **52** is a cross-sectional view taken along line I in FIG. **48**, shown without the internal components of the OPEP device;

[0083] FIG. **53** is a cross-sectional view taken along line I in FIG. **48**, shown with the internal components of the OPEP device;

[0084] FIG. **54** is a front-perspective view of an inner casing of the OPEP device of FIG. **48**;

[0085] FIG. **55** is a cross-sectional view of the inner casing taken along line I of in FIG. **54**;

[0086] FIG. **56** is a perspective view of a vane of the OPEP device of FIG. **48**;

[0087] FIG. **57** is a front perspective view of a restrictor member of the OPEP device of FIG. **48**;

[0088] FIG. **58** is a rear perspective view of the restrictor member of the FIG. **57**;

[0089] FIG. **59** is a front view of the restrictor member of FIG. **57**;

[0090] FIG. **60** is a front perspective view of an adjustment mechanism of the OPEP device of FIG. **48**;

[0091] FIG. **61** is a rear perspective view of the adjustment mechanism of FIG. **60**;

[0092] FIG. **62** is a front perspective view of the adjustment mechanism of FIGS. **60-61** assembled with the restrictor member of FIGS. **57-59** and the vane of FIG. **56**;

[0093] FIG. **63** is a front perspective view of a variable nozzle of the OPEP device of FIG. **48**;

[0094] FIG. **64** is a rear perspective view of the variable nozzle of FIG. **63**;

[0095] FIG. **65** is a front perspective view of the one-way valve of the OPEP device of FIG. **48**;

[0096] FIG. **66** is a perspective view of another embodiment of a respiratory treatment device;

[0097] FIG. **67** is an exploded view of the respiratory treatment device of FIG. **66**;

[0098] FIG. **68** is a cross-sectional perspective view taken along line I in FIG. **66** of the respiratory treatment device shown with the internal components of the device;

[0099] FIG. **69** is a cross-sectional perspective view taken along line II in FIG. **66** of the respiratory treatment device shown with the internal components of the device;

[0100] FIG. **70** is a different cross-sectional perspective view taken along line I in FIG. **66** of the respiratory treatment device, showing a portion of an exemplary exhalation flow path;

[0101] FIG. **71** is a different cross-sectional perspective view taken along line II in FIG. **66**,

showing a portion of an exemplary exhalation flow path;

[0102] FIG. 72 is another cross-sectional perspective view taken along line I in FIG. 66, showing a portion of an exemplary inhalation flow path;

[0103] FIG. 73 is another cross-sectional perspective view taken along line II in FIG. 66, showing a portion of an exemplary inhalation flow path; and,

[0104] FIG. 74 is a front perspective view of the OPEP device of FIG. 48, connected with an exemplary aerosol delivery device in the form of a nebulizer.

DETAILED DESCRIPTION

[0105] OPEP therapy is effective within a range of operating conditions. For example, an adult human may have an exhalation flow rate ranging from 10 to 60 liters per minute, and may maintain a static exhalation pressure in the range of 8 to 18 cm H₂O. Within these parameters, OPEP therapy is believed to be most effective when changes in the exhalation pressure (i.e., the amplitude) range from 5 to 20 cm H₂O oscillating at a frequency of 10 to 40 Hz. In contrast, an adolescent may have a much lower exhalation flow rate, and may maintain a lower static exhalation pressure, thereby altering the operating conditions most effective for the administration of OPEP therapy. Likewise, the ideal operating conditions for someone suffering from a respiratory illness, or in contrast, a healthy athlete, may differ from those of an average adult. As described below, the components of the disclosed OPEP devices are selectable and/or adjustable so that ideal operating conditions (e.g., amplitude and frequency of oscillating pressure) may be identified and maintained. Each of the various embodiments described herein achieve frequency and amplitude ranges that fall within the desired ranges set forth above. Each of the various embodiments described herein may also be configured to achieve frequencies and amplitudes that fall outside the ranges set forth above.

First Embodiment

[0106] Referring first to FIGS. 1-4, a front perspective view, a rear perspective view, a cross-sectional front perspective view, and an exploded view of an OPEP device 100 are shown. For purposes of illustration, the internal components of the OPEP device 100 are omitted in FIG. 3. The OPEP device 100 generally comprises a housing 102, a chamber inlet 104, a first chamber outlet 106, a second chamber outlet 108 (best seen in FIGS. 2 and 7), and a mouthpiece 109 in fluid communication with the chamber inlet 104. While the mouthpiece 109 is shown in FIGS. 1-4 as being integrally formed with the housing 102, it is envisioned that the mouthpiece 109 may be removable and replaceable with a mouthpiece 109 of a different size or shape, as required to maintain ideal operating conditions. In general, the housing 102 and the mouthpiece 109 may be constructed of any durable material, such as a polymer. One such material is Polypropylene. Alternatively, acrylonitrile butadiene styrene (ABS) may be used.

[0107] Alternatively, other or additional interfaces, such as breathing tubes or gas masks (not shown) may be attached in fluid communication with the mouthpiece 109 and/or associated with the housing 102. For example, the housing 102 may include an inhalation port (not shown) having a separate one-way inhalation valve (not shown) in fluid communication with the mouthpiece 109 to permit a user of the OPEP device 100 both to inhale the surrounding air through the one-way valve, and to exhale through the chamber inlet 104 without withdrawing the mouthpiece 109 of the OPEP device 100 between periods of inhalation and exhalation. In addition, any number of aerosol delivery devices may be connected to the OPEP device 100, for example, through the inhalation port mentioned above, for the simultaneous administration of aerosol and OPEP therapies. As such, the inhalation port may include, for example, an elastomeric adapter, or other flexible adapter, capable of accommodating the different mouthpieces or outlets of the particular aerosol delivery device that a user intends to use with the OPEP device 100. As used herein, the term aerosol delivery devices should be understood to include, for example, without limitation, any nebulizer, soft mist inhaler, pressurized metered dose inhaler, dry powder inhaler, combination of a holding chamber a pressurized metered dose inhaler, or the like. Suitable commercially available aerosol

delivery devices include, without limitation, the AEROECLIPSE nebulizer, RESPIMAT soft mist inhaler, LC Sprint nebulizer, AEROCHAMBER PLUS holding chambers, MICRO MIST nebulizer, SIDESTREAM nebulizers, Inspiration Elite nebulizers, FLOVENT pMDI, VENTOLIN pMDI, AZMACORT pMDI, BECLOVENT pMDI, QVAR pMDI and AEROBID pMDI, XOPENEX pMDI, PROAIR pMDI, PROVENT pMDI, SYMBICORT pMDI, TURBOHALER DPI, and DISKHALER DPI. Descriptions of suitable aerosol delivery devices may be found in U.S. Pat. Nos. 4,566,452; 5,012,803; 5,012,804; 5,312,046; 5,497,944; 5,622,162; 5,823,179; 6,293,279; 6,435,177; 6,484,717; 6,848,443; 7,360,537; 7,568,480; and, 7,905,228, the entireties of which are herein incorporated by reference.

[0108] In FIGS. 1-4, the housing **102** is generally box-shaped. However, a housing **102** of any shape may be used. Furthermore, the chamber inlet **104**, the first chamber outlet **106**, and the second chamber outlet **108** could be any shape or series of shapes, such as a plurality (i.e., more than one) of circular passages or linear slots. More importantly, it should be appreciated that the cross-sectional area of the chamber inlet **104**, the first chamber outlet **106**, and the second chamber outlet **108** are only a few of the factors influencing the ideal operating conditions described above.

[0109] Preferably, the housing **102** is openable so that the components contained therein can be periodically accessed, cleaned, replaced, or reconfigured, as required to maintain the ideal operating conditions. As such, the housing **102** is shown in FIGS. 1-4 as comprising a front section **101**, a middle section **103**, and a rear section **105**. The front section **101**, the middle section **103**, and the rear section **105** may be removably connected to one another by any suitable means, such as a snap-fit, a compression fit, etc., such that a seal forms between the relative sections sufficient to permit the OPEP device **100** to properly administer OPEP therapy.

[0110] As shown in FIG. 3, an exhalation flow path **110**, identified by a dashed line, is defined between the mouthpiece **109** and at least one of the first chamber outlet **106** and the second chamber outlet **108** (best seen in FIG. 7). More specifically, the exhalation flow path **110** begins at the mouthpiece **109**, passes through the chamber inlet **104**, and enters into a first chamber **114**, or an entry chamber. In the first chamber **114**, the exhalation flow path makes a 180-degree turn, passes through a chamber passage **116**, and enters into a second chamber **118**, or an exit chamber. In the second chamber **118**, the exhalation flow path **110** may exit the OPEP device **100** through at least one of the first chamber outlet **106** and the second chamber outlet **108**. In this way, the exhalation flow path **110** is “folded” upon itself, i.e., it reverses longitudinal directions between the chamber inlet **104** and one of the first chamber outlet **106** or the second chamber outlet **108**. However, those skilled in the art will appreciate that the exhalation flow path **110** identified by the dashed line is exemplary, and that air exhaled into the OPEP device **100** may flow in any number of directions or paths as it traverses from the mouthpiece **109** or chamber inlet **104** and the first chamber outlet **106** or the second chamber outlet **108**.

[0111] FIG. 3 also shows various other features of the OPEP device **100** associated with the housing **102**. For example, a stop **122** prevents a restrictor member **130** (see FIG. 5), described below, from opening in a wrong direction; a seat **124** shaped to accommodate the restrictor member **130** is formed about the chamber inlet **104**; and, an upper bearing **126** and a lower bearing **128** are formed within the housing **102** and configured to accommodate a shaft rotatably mounted therebetween. One or more guide walls **120** are positioned in the second chamber **118** to direct exhaled air along the exhalation flow path **110**.

[0112] Turning to FIGS. 5-7, various cross-sectional perspective views of the OPEP device **100** are shown with its internal components. The internal components of the OPEP device **100** comprise a restrictor member **130**, a vane **132**, and an optional variable nozzle **136**. As shown, the restrictor member **130** and the vane **132** are operatively connected by means of a shaft **134** rotatably mounted between the upper bearing **126** and the lower bearing **128**, such that the restrictor member **130** and the vane **132** are rotatable in unison about the shaft **134**. As described below in further detail, the variable nozzle **136** includes an orifice **138** configured to increase in size in response to the flow of

exhaled air therethrough.

[0113] FIGS. **4-6** further illustrate the division of the first chamber **114** and the second chamber **118** within the housing **102**. As previously described, the chamber inlet **104** defines an entrance to the first chamber **114**. The restrictor member **130** is positioned in the first chamber **114** relative to a seat **124** about the chamber inlet **104** such that it is moveable between a closed position, where a flow of exhaled air along the exhalation flow path **110** through the chamber inlet **104** is restricted, and an open position, where the flow of exhaled air through the chamber inlet **104** is less restricted. Likewise, the variable nozzle **136**, which is optional, is mounted about or positioned in the chamber passage **116**, such that the flow of exhaled air entering the first chamber **114** exits the first chamber **114** through the orifice **138** of the variable nozzle **136**. Exhaled air exiting the first chamber **114** through the orifice **138** of the variable nozzle **136** enters the second chamber, which is defined by the space within the housing **102** occupied by the vane **132** and the guide walls **120**. Depending on the position of the vane **132**, the exhaled air is then able to exit the second chamber **118** through at least one of the first chamber outlet **106** and the second chamber outlet **108**.

[0114] FIGS. **8-14** show the internal components of the OPEP device **100** in greater detail. Turning first to FIGS. **8-9**, a front perspective view and a rear perspective view shows the restrictor member **130** operatively connected to the vane **132** by the shaft **134**. As such, the restrictor member **130** and the vane **132** are rotatable about the shaft **134** such that rotation of the restrictor member **130** results in a corresponding rotation of the vane **132**, and vice-versa. Like the housing **102**, the restrictor member **130** and the vane **132** may be made of constructed of any durable material, such as a polymer. Preferably, they are constructed of a low shrink, low friction plastic. One such material is acetal.

[0115] As shown, the restrictor member **130**, the vane **132**, and the shaft **134** are formed as a unitary component. The restrictor member **130** is generally disk-shaped, and the vane **132** is planar. The restrictor member **130** includes a generally circular face **140** axially offset from the shaft **134** and a beveled or chamfered edge **142** shaped to engage the seat **124** formed about the chamber inlet **104**. In this way, the restrictor member **130** is adapted to move relative to the chamber inlet **104** about an axis of rotation defined by the shaft **134** such that the restrictor member **130** may engage the seat **124** in a closed position to substantially seal and restrict the flow of exhaled air through the chamber inlet **104**. However, it is envisioned that the restrictor member **130** and the vane **132** may be formed as separate components connectable by any suitable means such that they remain independently replaceable with a restrictor member **130** or a vane **132** of a different shape, size, or weight, as selected to maintain ideal operating conditions. For example, the restrictor member **130** and/or the vane **132** may include one or more contoured surfaces. Alternatively, the restrictor member **130** may be configured as a butterfly valve.

[0116] Turning to FIG. **10**, a front view of the restrictor member **130** and the vane **132** is shown. As previously described, the restrictor member **130** comprises a generally circular face **140** axially offset from the shaft **134**. The restrictor member **130** further comprises a second offset designed to facilitate movement of the restrictor member **130** between a closed position and an open position. More specifically, a center **144** of the face **140** of the restrictor member **130** is offset from the plane defined by the radial offset and the shaft **134**, or the axis of rotation. In other words, a greater surface area of the face **140** of the restrictor member **130** is positioned on one side of the shaft **134** than on the other side of the shaft **134**. Pressure at the chamber inlet **104** derived from exhaled air produces a force acting on the face **140** of the restrictor member **130**. Because the center **144** of the face **140** of the restrictor member **130** is offset as described above, a resulting force differential creates a torque about the shaft **134**. As further explained below, this torque facilitates movement of the restrictor member **130** between a closed position and an open position.

[0117] Turning to FIG. **11**, a top view of the restrictor member **130** and the vane **132** is shown. As illustrated, the vane **132** is connected to the shaft **134** at a 75° angle relative to the face **140** of restrictor member **130**. Preferably, the angle will remain between 60° and 80°, although it is

envisioned that the angle of the vane **132** may be selectively adjusted to maintain the ideal operating conditions, as previously discussed. It is also preferable that the vane **132** and the restrictor member **130** are configured such that when the OPEP device **100** is fully assembled, the angle between a centerline of the variable nozzle **136** and the vane **132** is between 10° and 25° when the restrictor member **130** is in a closed position. Moreover, regardless of the configuration, it is preferable that the combination of the restrictor member **130** and the vane **132** have a center of gravity aligned with the shaft **134**, or the axis of rotation. In full view of the present disclosure, it should be apparent to those skilled in the art that the angle of the vane **132** may be limited by the size or shape of the housing **102**, and will generally be less than half the total rotation of the vane **132** and the restrictor member **130**.

[0118] Turning to FIGS. **12** and **13**, a front perspective view and a rear perspective view of the variable nozzle **136** is shown without the flow of exhaled air therethrough. In general, the variable nozzle **136** includes top and bottom walls **146**, side walls **148**, and V-shaped slits **150** formed therebetween. As shown, the variable nozzle is generally shaped like a duck-bill type valve. However, it should be appreciated that nozzles or valves of other shapes and sizes may also be used. The variable nozzle **136** may also include a lip **152** configured to mount the variable nozzle **136** within the housing **102** between the first chamber **114** and the second chamber **118**. The variable nozzle **136** may be constructed or molded of any material having a suitable flexibility, such as silicone, and preferably with a wall thickness of between 0.50 and 2.00 millimeters, and an orifice width between 0.25 to 1.00 millimeters, or smaller depending on manufacturing capabilities. [0119] As previously described, the variable nozzle **136** is optional in the operation of the OPEP device **100**. It should also be appreciated that the OPEP device **100** could alternatively omit both the chamber passage **116** and the variable nozzle **136**, and thus comprise a single-chamber embodiment. Although functional without the variable nozzle **136**, the performance of the OPEP device **100** over a wider range of exhalation flow rates is improved when the OPEP device **100** is operated with the variable nozzle **136**. The chamber passage **116**, when used without the variable nozzle **136**, or the orifice **138** of the variable nozzle **136**, when the variable nozzle **136** is included, serves to create a jet of exhaled air having an increased velocity. As explained in more detail below, the increased velocity of the exhaled air entering the second chamber **118** results in a proportional increase in the force applied by the exhaled air to the vane **132**, and in turn, an increased torque about the shaft **134**, all of which affect the ideal operating conditions.

[0120] Without the variable nozzle **136**, the orifice between the first chamber **114** and the second chamber **118** is fixed according to the size, shape, and cross-sectional area of the chamber passage **116**, which may be selectively adjusted by any suitable means, such as replacement of the middle section **103** or the rear section **105** of the housing. On the other hand, when the variable nozzle **136** is included in the OPEP device **100**, the orifice between the first chamber **114** and the second chamber **118** is defined by the size, shape, and cross-sectional area of the orifice **138** of the variable nozzle **136**, which may vary according to the flow rate of exhaled air and/or the pressure in the first chamber **114**.

[0121] Turning to FIG. **14**, a front perspective view of the variable nozzle **136** is shown with a flow of exhaled air therethrough. One aspect of the variable nozzle **136** shown in FIG. **14** is that, as the orifice **138** opens in response to the flow of exhaled air therethrough, the cross-sectional shape of the orifice **138** remains generally rectangular, which during the administration of OPEP therapy results in a lower drop in pressure through the variable nozzle **136** from the first chamber **114** (See FIGS. **3** and **5**) to the second chamber **118**. The generally consistent rectangular shape of the orifice **138** of the variable nozzle **136** during increased flow rates is achieved by the V-shaped slits **150** formed between the top and bottom walls **146** and the side walls **148**, which serve to permit the side walls **148** to flex without restriction. Preferably, the V-shaped slits **150** are as thin as possible to minimize the leakage of exhaled air therethrough. For example, the V-shaped slits **150** may be approximately 0.25 millimeters wide, but depending on manufacturing capabilities, could range

between 0.10 and 0.50 millimeters. Exhaled air that does leak through the V-shaped slits **150** is ultimately directed along the exhalation flow path by the guide walls **120** in the second chamber **118** protruding from the housing **102**.

[0122] It should be appreciated that numerous factors contribute to the impact the variable nozzle **136** has on the performance of the OPEP device **100**, including the geometry and material of the variable nozzle **136**. By way of example only, in order to attain a target oscillating pressure frequency of between 10 to 13 Hz at an exhalation flow rate of 15 liters per minute, in one embodiment, a 1.0 by 20.0 millimeter passage or orifice may be utilized. However, as the exhalation flow rate increases, the frequency of the oscillating pressure in that embodiment also increases, though at a rate too quickly in comparison to the target frequency. In order to attain a target oscillating pressure frequency of between 18 to 20 Hz at an exhalation flow rate of 45 liters per minute, the same embodiment may utilize a 3.0 by 20.0 millimeter passage or orifice. Such a relationship demonstrates the desirability of a passage or orifice that expands in cross-sectional area as the exhalation flow rate increases in order to limit the drop in pressure across the variable nozzle **136**.

[0123] Turning to FIGS. **15A-C**, top phantom views of the OPEP device **100** show an exemplary illustration of the operation of the OPEP device **100**. Specifically, FIG. **15A** shows the restrictor member **130** in an initial, or closed position, where the flow of exhaled air through the chamber inlet **104** is restricted, and the vane **132** is in a first position, directing the flow of exhaled air toward the first chamber outlet **106**. FIG. **15B** shows this restrictor member **130** in a partially open position, where the flow of exhaled air through the chamber inlet **104** is less restricted, and the vane **132** is directly aligned with the jet of exhaled air exiting the variable nozzle **136**. FIG. **15C** shows the restrictor member **130** in an open position, where the flow of exhaled air through the chamber inlet **104** is even less restricted, and the vane **132** is in a second position, directing the flow of exhaled air toward the second chamber outlet **108**. It should be appreciated that the cycle described below is merely exemplary of the operation of the OPEP device **100**, and that numerous factors may affect operation of the OPEP device **100** in a manner that results in a deviation from the described cycle. However, during the operation of the OPEP device **100**, the restrictor member **130** and the vane **132** will generally reciprocate between the positions shown in FIGS. **15A** and **15C**.

[0124] During the administration of OPEP therapy, the restrictor member **130** and the vane **132** may be initially positioned as shown in FIG. **15A**. In this position, the restrictor member **130** is in a closed position, where the flow of exhaled air along the exhalation path through the chamber inlet **104** is substantially restricted. As such, an exhalation pressure at the chamber inlet **104** begins to increase when a user exhales into the mouthpiece **108**. As the exhalation pressure at the chamber inlet **104** increases, a corresponding force acting on the face **140** of the restrictor member **130** increases. As previously explained, because the center **144** of the face **140** is offset from the plane defined by the radial offset and the shaft **134**, a resulting net force creates a negative or opening torque about the shaft. In turn, the opening torque biases the restrictor member **130** to rotate open, letting exhaled air enter the first chamber **114**, and biases the vane **132** away from its first position. As the restrictor member **130** opens and exhaled air is let into the first chamber **114**, the pressure at the chamber inlet **104** begins to decrease, the force acting on the face **140** of the restrictor member begins to decrease, and the torque biasing the restrictor member **130** open begins to decrease.

[0125] As exhaled air continues to enter the first chamber **114** through the chamber inlet **104**, it is directed along the exhalation flow path **110** by the housing **102** until it reaches the chamber passage **116** disposed between the first chamber **114** and the second chamber **118**. If the OPEP device **100** is being operated without the variable nozzle **136**, the exhaled air accelerates through the chamber passage **116** due to the decrease in cross-sectional area to form a jet of exhaled air. Likewise, if the OPEP device **100** is being operated with the variable nozzle **136**, the exhaled air accelerates through the orifice **138** of the variable nozzle **136**, where the pressure through the orifice **138**

causes the side walls **148** of the variable nozzle **136** to flex outward, thereby increasing the size of the orifice **138**, as well as the resulting flow of exhaled air therethrough. To the extent some exhaled air leaks out of the V-shaped slits **150** of the variable nozzle **136**, it is directed back toward the jet of exhaled air and along the exhalation flow path by the guide walls **120** protruding into the housing **102**.

[0126] Then, as the exhaled air exits the first chamber **114** through the variable nozzle **136** and/or chamber passage **116** and enters the second chamber **118**, it is directed by the vane **132** toward the front section **101** of the housing **102**, where it is forced to reverse directions before exiting the OPEP device **100** through the open first chamber exit **106**. As a result of the change in direction of the exhaled air toward the front section **101** of the housing **102**, a pressure accumulates in the second chamber **118** near the front section **101** of the housing **102**, thereby resulting in a force on the adjacent vane **132**, and creating an additional negative or opening torque about the shaft **134**. The combined opening torques created about the shaft **134** from the forces acting on the face **140** of the restrictor member **130** and the vane **132** cause the restrictor member **130** and the vane **132** to rotate about the shaft **134** from the position shown in FIG. **15A** toward the position shown in FIG. **15B**.

[0127] When the restrictor member **130** and the vane **132** rotate to the position shown in FIG. **15B**, the vane **132** crosses the jet of exhaled air exiting the variable nozzle **136** or the chamber passage **116**. Initially, the jet of exhaled air exiting the variable nozzle **136** or chamber passage **116** provides a force on the vane **132** that, along with the momentum of the vane **132**, the shaft **134**, and the restrictor member **130**, propels the vane **132** and the restrictor member **130** to the position shown in FIG. **15C**. However, around the position shown in FIG. **15B**, the force acting on the vane **132** from the exhaled air exiting the variable nozzle **136** also switches from a negative or opening torque to a positive or closing torque. More specifically, as the exhaled air exits the first chamber **114** through the variable nozzle **136** and enters the second chamber **118**, it is directed by the vane **132** toward the front section **101** of the housing **102**, where it is forced to reverse directions before exiting the OPEP device **100** through the open second chamber exit **108**. As a result of the change in direction of the exhaled air toward the front section **101** of the housing **102**, a pressure accumulates in the second chamber **118** near the front section **101** of the housing **102**, thereby resulting in a force on the adjacent vane **132**, and creating a positive or closing torque about the shaft **134**. As the vane **132** and the restrictor member **130** continue to move closer to the position shown in FIG. **15C**, the pressure accumulating in the section chamber **118** near the front section **101** of the housing **102**, and in turn, the positive or closing torque about the shaft **134**, continues to increase, as the flow of exhaled air along the exhalation flow path **110** and through the chamber inlet **104** is even less restricted. Meanwhile, although the torque about the shaft **134** from the force acting on the restrictor member **130** also switches from a negative or opening torque to a positive or closing torque around the position shown in FIG. **15B**, its magnitude is essentially negligible as the restrictor member **130** and the vane **132** rotate from the position shown in FIG. **15B** to the position shown in FIG. **15C**.

[0128] After reaching the position shown in FIG. **15C**, and due to the increased positive or closing torque about the shaft **134**, the vane **132** and the restrictor member **130** reverse directions and begin to rotate back toward the position shown in FIG. **15B**. As the vane **132** and the restrictor member **130** approach the position shown in FIG. **15B**, and the flow of exhaled through the chamber inlet **104** is increasingly restricted, the positive or closing torque about the shaft **134** begins to decrease. When the restrictor member **130** and the vane **132** reach the position **130** shown in FIG. **15B**, the vane **132** crosses the jet of exhaled air exiting the variable nozzle **136** or the chamber passage **116**, thereby creating a force on the vane **132** that, along with the momentum of the vane **132**, the shaft **134**, and the restrictor member **130**, propels the vane **132** and the restrictor member **130** back to the position shown in FIG. **15A**. After the restrictor member **130** and the vane **132** return to the position shown in FIG. **15A**, the flow of exhaled air through the chamber inlet **104** is restricted, and

the cycle described above repeats itself.

[0129] It should be appreciated that, during a single period of exhalation, the cycle described above will repeat numerous times. Thus, by repeatedly moving the restrictor member **130** between a closed position, where the flow of exhaled air through the chamber inlet **104** is restricted, and an open position, where the flow of exhaled air through the chamber inlet **104** is less restricted, an oscillating back pressure is transmitted to the user of the OPEP device **100** and OPEP therapy is administered.

[0130] Turning now to FIGS. **16-17**, an alternative embodiment of a variable nozzle **236** is shown. The variable nozzle **236** may be used in the OPEP device **100** as an alternative to the variable nozzle **136** described above. As shown in FIGS. **16-17**, the variable nozzle **236** includes an orifice **238**, top and bottom walls **246**, side walls **248**, and a lip **252** configured to mount the variable nozzle **236** within the housing of the OPEP device **100** between the first chamber **114** and the second chamber **118** in the same manner as the variable nozzle **136**. Similar to the variable nozzle **136** shown in FIGS. **12-13**, the variable nozzle **236** may be constructed or molded of any material having a suitable flexibility, such as silicone.

[0131] During the administration of OPEP therapy, as the orifice **238** of the variable nozzle **236** opens in response to the flow of exhaled air therethrough, the cross-sectional shape of the orifice **238** remains generally rectangular, which results in a lower drop in pressure through the variable nozzle **236** from the first chamber **114** to the second chamber **118**. The generally consistent rectangular shape of the orifice **238** of the variable nozzle **236** during increased flow rates is achieved by thin, creased walls formed in the top and bottom walls **246**, which allow the side walls **248** to flex easier and with less resistance. A further advantage of this embodiment is that there is no leakage out of the top and bottom walls **246** while exhaled air flows through the orifice **238** of the variable nozzle **236**, such as for example, through the V-shaped slits **150** of the variable nozzle **136** shown in FIGS. **12-13**.

[0132] Those skilled in the art will also appreciate that, in some applications, only positive expiratory pressure (without oscillation) may be desired, in which case the OPEP device **100** may be operated without the restrictor member **130**, but with a fixed orifice or manually adjustable orifice instead. The positive expiratory pressure embodiment may also comprise the variable nozzle **136**, or the variable nozzle **236**, in order to maintain a relatively consistent back pressure within a desired range.

Second Embodiment

[0133] Turning now to FIGS. **18-19**, a front perspective view and a rear perspective view of a second embodiment of an OPEP device **200** is shown. The configuration and operation of the OPEP device **200** is similar to that of the OPEP device **100**. However, as best shown in FIGS. **20-24**, the OPEP device **200** further includes an adjustment mechanism **253** adapted to change the relative position of the chamber inlet **204** with respect to the housing **202** and the restrictor member **230**, which in turn changes the range of rotation of the vane **232** operatively connected thereto. As explained below, a user is therefore able to conveniently adjust both the frequency and the amplitude of the OPEP therapy administered by the OPEP device **200** without opening the housing **202** and disassembling the components of the OPEP device **200**.

[0134] The OPEP device **200** generally comprises a housing **202**, a chamber inlet **204**, a first chamber outlet **206** (best seen in FIGS. **23** and **32**), a second chamber outlet **208** (best seen in FIGS. **23** and **32**), and a mouthpiece **209** in fluid communication with the chamber inlet **204**. As with the OPEP device **100**, a front section **201**, a middle section **203**, and a rear section **205** of the housing **202** are separable so that the components contained therein can be periodically accessed, cleaned, replaced, or reconfigured, as required to maintain the ideal operating conditions. The OPEP device also includes an adjustment dial **254**, as described below.

[0135] As discussed above in relation to the OPEP device **100**, the OPEP device **200** may be adapted for use with other or additional interfaces, such as an aerosol delivery device. In this

regard, the OPEP device **200** is equipped with an inhalation port **211** (best seen in FIGS. **19**, **21**, and **23**) in fluid communication with the mouthpiece **209** and the chamber inlet **204**. As noted above, the inhalation port may include a separate one-way valve (not shown) to permit a user of the OPEP device **200** both to inhale the surrounding air through the one-way valve and to exhale through the chamber inlet **204** without withdrawing the mouthpiece **209** of the OPEP device **200** between periods of inhalation and exhalation. In addition, the aforementioned aerosol delivery devices may be connected to the inhalation port **211** for the simultaneous administration of aerosol and OPEP therapies.

[0136] An exploded view of the OPEP device **200** is shown in FIG. **20**. In addition to the components of the housing described above, the OPEP device **200** includes a restrictor member **230** operatively connected to a vane **232** by a pin **231**, an adjustment mechanism **253**, and a variable nozzle **236**. As shown in the cross-sectional view of FIG. **21**, when the OPEP device **200** is in use, the variable nozzle **236** is positioned between the middle section **203** and the rear section **205** of the housing **202**, and the adjustment mechanism **253**, the restrictor member **230**, and the vane **232** form an assembly.

[0137] Turning to FIGS. **21-23**, various cross-sectional perspective views of the OPEP device **200** are shown. As with the OPEP device **100**, an exhalation flow path **210**, identified by a dashed line, is defined between the mouthpiece **209** and at least one of the first chamber outlet **206** and the second chamber outlet **208** (best seen in FIGS. **23** and **32**). As a result of a one-way valve (not-shown) and/or an aerosol delivery device (not shown) attached to the inhalation port **211**, the exhalation flow path **210** begins at the mouthpiece **209** and is directed toward the chamber inlet **204**, which in operation may or may not be blocked by the restrictor member **230**. After passing through the chamber inlet **204**, the exhalation flow path **210** enters a first chamber **214** and makes a 180° turn toward the variable nozzle **236**. After passing through the orifice **238** of the variable nozzle **236**, the exhalation flow path **210** enters a second chamber **218**. In the second chamber **218**, the exhalation flow path **210** may exit the OPEP device **200** through at least one of the first chamber outlet **206** or the second chamber outlet **208**. Those skilled in the art will appreciate that the exhalation flow path **210** identified by the dashed line is exemplary, and that air exhaled into the OPEP device **200** may flow in any number of directions or paths as it traverses from the mouthpiece **209** or chamber inlet **204** to the first chamber outlet **206** or the second chamber outlet **208**.

[0138] Referring to FIGS. **24-25**, front and rear perspective views of the adjustment mechanism **253** of the OPEP device **200** are shown. In general, the adjustment mechanism **253** includes an adjustment dial **254**, a shaft **255**, and a frame **256**. A protrusion **258** is positioned on a rear face **260** of the adjustment dial, and is adapted to limit the selective rotation of the adjustment mechanism **253** by a user, as further described below. The shaft **255** includes keyed portions **262** adapted to fit within upper and lower bearings **226**, **228** formed in the housing **200** (see FIGS. **21** and **28-29**). The shaft further includes an axial bore **264** configured to receive the pin **231** operatively connecting the restrictor member **230** and the vane **232**. As shown, the frame **256** is spherical, and as explained below, is configured to rotate relative to the housing **202**, while forming a seal between the housing **202** and the frame **256** sufficient to permit the administration of OPEP therapy. The frame **256** includes a circular opening defined by a seat **224** adapted to accommodate the restrictor member **230**. In use, the circular opening functions as the chamber inlet **204**. The frame **256** also includes a stop **222** for preventing the restrictor member **230** from opening in a wrong direction.

[0139] Turning to FIG. **26**, a front perspective view of the restrictor member **230** and the vane **232** is shown. The design, materials, and configuration of the restrictor member **230** and the vane **232** may be the same as described above in regards to the OPEP device **100**. However, the restrictor member **230** and the vane **232** in the OPEP device **200** are operatively connected by a pin **231** adapted for insertion through the axial bore **264** in the shaft **255** of the adjustment mechanism **253**. The pin **231** may be constructed, for example, by stainless steel. In this way, rotation of the

restrictor member **230** results in a corresponding rotation of the vane **232**, and vice versa.

[0140] Turning to FIG. **27**, a front perspective view of the adjustment mechanism **253** assembled with the restrictor member **230** and the vane **232** is shown. In this configuration, it can be seen that the restrictor member **230** is positioned such that it is rotatable relative to the frame **256** and the seat **224** between a closed position (as shown), where a flow of exhaled air along the exhalation flow path **210** through the chamber inlet **204** is restricted, and an open position (not shown), where the flow of exhaled air through the chamber inlet **204** is less restricted. As previously mentioned the vane **232** is operatively connected to the restrictor member **230** by the pin **231** extending through shaft **255**, and is adapted to move in unison with the restrictor member **230**. It can further be seen that the restrictor member **230** and the vane **232** are supported by the adjustment mechanism **253**, which itself is rotatable within the housing **202** of the OPEP device **200**, as explained below.

[0141] FIGS. **28** and **29A-B** are partial cross-sectional views illustrating the adjustment mechanism **253** mounted within the housing **202** of the OPEP device **200**. As shown in FIG. **28**, the adjustment mechanism **253**, as well as the restrictor member **230** and the vane **232**, are rotatably mounted within the housing **200** about an upper and lower bearing **226**, **228**, such that a user is able to rotate the adjustment mechanism **253** using the adjustment dial **254**. FIGS. **29A-29B** further illustrates the process of mounting and locking the adjustment mechanism **253** within the lower bearing **228** of the housing **202**. More specifically, the keyed portion **262** of the shaft **255** is aligned with and inserted through a rotational lock **166** formed in the housing **202**, as shown in FIG. **29A**. Once the keyed portion **262** of the shaft **255** is inserted through the rotational lock **266**, the shaft **255** is rotated 90° to a locked position, but remains free to rotate. The adjustment mechanism **253** is mounted and locked within the upper bearing **226** in the same manner.

[0142] Once the housing **200** and the internal components of the OPEP device **200** are assembled, the rotation of the shaft **255** is restricted to keep it within a locked position in the rotational lock **166**. As shown in a front view of the OPEP device **200** in FIG. **30**, two stops **268**, **288** are positioned on the housing **202** such that they engage the protrusion **258** formed on the rear face **260** of the adjustment dial **254** when a user rotates the adjustment dial **254** to a predetermined position. For purposes of illustration, the OPEP device **200** is shown in FIG. **30** without the adjustment dial **254** or the adjustment mechanism **253**, which would extend from the housing **202** through an opening **269**. In this way, rotation of the adjustment dial **254**, the adjustment mechanism **253**, and the keyed portion **262** of the shaft **255** can be appropriately restricted.

[0143] Turning to FIG. **31**, a partial cross-sectional view of the adjustment mechanism **253** mounted within the housing **200** is shown. As previously mentioned, the frame **256** of the adjustment mechanism **253** is spherical, and is configured to rotate relative to the housing **202**, while forming a seal between the housing **202** and the frame **256** sufficient to permit the administration of OPEP therapy. As shown in FIG. **31**, a flexible cylinder **271** extending from the housing **202** completely surrounds a portion of the frame **256** to form a sealing edge **270**. Like the housing **202** and the restrictor member **230**, the flexible cylinder **271** and the frame **256** may be constructed of a low shrink, low friction plastic. One such material is acetal. In this way, the sealing edge **270** contacts the frame **256** for a full 360° and forms a seal throughout the permissible rotation of the adjustment member **253**.

[0144] The selective adjustment of the OPEP device **200** will now be described with reference to FIGS. **32A-B**, **33A-B**, and **34A-B**. FIGS. **32A-B** are partial cross-sectional views of the OPEP device **200**; FIGS. **33A-B** are illustrations of the adjustability of the OPEP device **200**; and, FIGS. **34A-B** are top phantom views of the OPEP device **200**. As previously mentioned with regards to the OPEP device **100**, it is preferable that the vane **232** and the restrictor member **230** are configured such that when the OPEP device **200** is fully assembled, the angle between a centerline of the variable nozzle **236** and the vane **232** is between 10° and 25° when the restrictor member **230** is in a closed position. However, it should be appreciated that the adjustability of the OPEP

device **200** is not limited to the parameters described herein, and that any number of configurations may be selected for purposes of administering OPEP therapy within the ideal operating conditions. [0145] FIG. **32A** shows the vane **232** at an angle of 10° from the centerline of the variable nozzle **236**, whereas FIG. **32B** shows the vane **232** at an angle of 25° from the centerline of the variable nozzle **236**. FIG. **33A** illustrates the necessary position of the frame **256** (shown in phantom) relative to the variable nozzle **236** such that the angle between a centerline of the variable nozzle **236** and the vane **232** is 10° when the restrictor member **230** is in the closed position. FIG. **33B**, on the other hand, illustrates the necessary position of the frame **256** (shown in phantom) relative to the variable nozzle **236** such that the angle between a centerline of the variable nozzle **236** and the vane **232** is 25° when the restrictor member **230** is in the closed position.

[0146] Referring to FIGS. **34A-B**, side phantom views of the OPEP device **200** are shown. The configuration shown in FIG. **34A** corresponds to the illustrations shown in FIGS. **32A** and **33A**, wherein the angle between a centerline of the variable nozzle **236** and the vane **232** is 10° when the restrictor member **230** is in the closed position. FIG. **34B**, on the other hand, corresponds to the illustrations shown in FIGS. **32B** and **33B**, wherein the angle between a centerline of the variable nozzle **236** and the vane **232** is 25° when the restrictor member **230** is in the closed position. In other words, the frame **256** of the adjustment member **253** has been rotated counter-clockwise 15° , from the position shown in FIG. **34A**, to the position shown in FIG. **34B**, thereby also increasing the permissible rotation of the vane **232**.

[0147] In this way, a user is able to rotate the adjustment dial **254** to selectively adjust the orientation of the chamber inlet **204** relative to the restrictor member **230** and the housing **202**. For example, a user may increase the frequency and amplitude of the OPEP therapy administered by the OPEP device **200** by rotating the adjustment dial **254**, and therefore the frame **256**, toward the position shown in FIG. **34A**. Alternatively, a user may decrease the frequency and amplitude of the OPEP therapy administered by the OPEP device **200** by rotating the adjustment dial **254**, and therefore the frame **256**, toward the position shown in FIG. **34B**. Furthermore, as shown for example in FIGS. **18** and **30**, indicia may be provided to aid the user in the setting of the appropriate configuration of the OPEP device **200**.

[0148] Operating conditions similar to those described below with reference to the OPEP device **800** may also be achievable for an OPEP device according to the OPEP device **200**.

Third Embodiment

[0149] Turning now to FIGS. **35-38**, a third embodiment of an OPEP device **300** is shown. As described below, with the exception of an adjustment mechanism **353**, the design and operation of the OPEP device **300** is the same as the OPEP device **200**. For example, as seen in the front perspective view of FIG. **35**, a housing **302** of the OPEP device **300** includes a mouthpiece **309**, a first chamber outlet **306**, and a second chamber outlet (not shown) positioned opposite the first chamber outlet **306**. The housing **302** is formed of a front section **301**, a middle section **303**, and a rear section **305**. As shown in the cross-sectional view of FIG. **36**, the OPEP device **300** also includes a restrictor member **330** operatively connected to a vane **332** by a shaft (not shown), and a variable nozzle **336** separating a first chamber **314** and a second chamber **318**. Finally, an exhalation flow path **310**, identified by a dashed line, is formed between the mouthpiece **309** and at least one of the first chamber outlet **306** and the second chamber outlet. Those skilled in the art will appreciate that the exhalation flow path **310** identified by the dashed line is exemplary, and that air exhaled into the OPEP device **300** may flow in any number of directions or paths as it traverses from the mouthpiece **309** or chamber inlet **304** to the first chamber outlet **306** or the second chamber outlet.

[0150] Referring to FIGS. **37-38**, a front perspective view and a rear perspective view of the adjustment mechanism **353** assembled with the restrictor member **330** and the vane **332** are shown. The adjustment mechanism **353** is comprised of a cup **372** shaped to fit within the housing **302** such that a user may rotate the cup **372** relative to the housing **302** via an adjustment dial **354**. A

wall **374** extends through the central portion of the cup **372**. The wall **372** includes an opening defined by a seat **324** shaped to accommodate the restrictor member **330**. As seen in FIG. **36**, the opening operates as the chamber inlet **304** during the administration of OPEP therapy. The cup **372** further includes an upper bearing **326** and a lower bearing **328** adapted to rotatably mount the restrictor member **330**, the vane **332**, and the shaft (not shown) to the adjustment mechanism **353**, such that the restrictor member **330** and the vane **332** are rotatable relative to the cup **372**. The wall also includes a stop **322** to prevent the restrictor member **330** from opening in a wrong direction. [0151] When the OPEP device **300** is fully assembled as shown in FIGS. **35-36**, a user is able to rotate the adjustment dial **354** relative to the housing **302** to selectively adjust the frequency and amplitude of the OPEP therapy administered by the OPEP device **300**. Similar to the adjustment mechanism **253** of the OPEP device **200**, a user may adjust the orientation of the chamber inlet **304** relative to the restrictor member **330** in the OPEP device **300** by rotating the adjustment dial **354**, thereby rotating the cup **372** and the wall **374** relative to the restrictor member **330** and the housing **302**. A user may increase the frequency and amplitude of the OPEP therapy administered by the OPEP device **300** by rotating the adjustment dial **354**, and therefore the wall **374**, in the clockwise direction. Alternatively, a user may decrease the frequency and amplitude of the OPEP therapy administered by the OPEP device **300** by rotating the adjustment dial **354**, and therefore the wall **374**, in the counter-clockwise direction. As shown in FIGS. **35-36**, a protrusion **358** extending from the housing **302** through a slot **376** in the adjustment dial **354** may be provided to restrict the rotation of the adjustment dial **354** such that the permissible configurations of the OPEP device **300** are limited, and the ideal operating conditions are maintained.

[0152] Operating conditions similar to those described below with reference to the OPEP device **800** may also be achievable for an OPEP device according to the OPEP device **300**.

Fourth Embodiment

[0153] Turning to FIGS. **39-40**, a fourth embodiment of an OPEP device **400** is shown. Although the configuration of the OPEP device **400** differs from that of the OPEP device **300** and the OPEP device **200**, the internal components and operation of the OPEP device **400** are otherwise the same. For example, as seen in the front perspective view of FIG. **39**, a housing **402** of the OPEP device **400** includes a mouthpiece **409**, a first chamber outlet **406**, and a second chamber outlet **408** (best seen in FIGS. **41A-41B**) positioned opposite the first chamber outlet. The housing **402** is formed of a front section **401**, a middle section **403**, and a rear section **405**, as well as an upper section **407** adapted to rotate relative to the front section **401**, the middle section **403**, and the rear section **405**.

[0154] As seen in the cross-sectional view of FIG. **40**, the OPEP device **400** further includes a restrictor member **430** operatively connected to a vane **432** by a shaft (not shown), and a variable nozzle **436** separating a first chamber **414** and a second chamber **418**. The upper section **407** of the housing **402** includes a frame **456** having a seat **424** shaped to accommodate the restrictor member **430**, a stop **422** to prevent the restrictor member **430** from opening in a wrong direction, as well as an upper bearing **426** and a lower bearing **428** about which the shaft (not shown) operatively connecting the restrictor member **430** and the vane **436** is rotatably mounted. In operation, the seat **422** defines the chamber inlet **404**. Consequently, the restrictor member **430** is rotatable relative to the seat **422** and the chamber inlet **404**.

[0155] As with the previously described embodiments, an exhalation flow path **410**, identified by a dashed line, is formed between the mouthpiece **409** and at least one of the first chamber outlet **406** and the second chamber outlet **408**. Once again, those skilled in the art will appreciate that the exhalation flow path **410** identified by the dashed line is exemplary, and that air exhaled into the OPEP device **400** may flow in any number of directions or paths as it traverses from the mouthpiece **409** or chamber inlet **404** to the first chamber outlet **406** or the second chamber outlet **408**. Due to the configuration of the OPEP device **400**, the exhalation flow path **410** differs from those of the embodiments described above. More specifically, the exhalation flow path **410** begins at the mouthpiece **409** formed in the upper section **407** of the housing **402**, passes through the

chamber inlet **404**, and enters into a first chamber **114**. In the first chamber **414**, the exhalation flow path makes a 180° turn in the direction of the front section **401** of the housing **402**, followed by a 90° turn toward the bottom of the OPEP device **400**, past a second chamber **418** of the housing **402**. The exhalation flow path **410** then makes a 90° turn toward the rear section **405** of the housing **402**, where it makes another 180° turn and passes through a variable nozzle **436**, and enters into the second chamber **418**. In the second chamber **418**, the exhalation flow path **410** may exit the OPEP device **410** through at least one of the first chamber outlet **406** or the second chamber outlet **408**. [0156] As seen in FIGS. **40** and **41A-B**, the upper section **407** of the housing **402** is rotatable relative to the front section **401**, the middle section **403**, and the rear section **405** of the housing **402**. In this way, a user is able to rotate the upper section **407** relative to the front section **401**, the middle section **403**, and the rear section **405** to selectively adjust the orientation of the chamber inlet **404** relative to the restrictor member **430** and the housing **402**, and thereby selectively adjust the frequency and amplitude of the OPEP therapy administered by the OPEP device **400**, in a similar manner as previously described in relation to the adjustability of the OPEP device **200**. For example, a user may increase the frequency and amplitude of the OPEP therapy administered by the OPEP device **400** by rotating the upper section **407**, and therefore the seat **422**, relative to the front section **401**, the middle section **403**, and the rear section **405**, toward the position shown in FIG. **41A**. Alternatively, a user may decrease the frequency and amplitude of the OPEP therapy administered by the OPEP device **400** by rotating the upper section **407**, and therefore the seat **422**, relative to the front section **401**, the middle section **403**, and the rear section **405**, toward the position shown in FIG. **41B**. Furthermore, as shown in FIGS. **40** and **41A-B**, a protrusion **458** extending from the middle section **403** of the housing **402** may be provided to restrict the rotation of the upper section **407** such that the permissible configurations of the OPEP device **400** are limited, and the ideal operating conditions are maintained.

[0157] Operating conditions similar to those described below with reference to the OPEP device **800** may also be achievable for an OPEP device according to the OPEP device **400**.

[0158] Turning to FIGS. **42-47**, various alternative embodiments of the OPEP device **100** are shown. Although the embodiments shown in FIGS. **42-47** and described below are alternative embodiments of the OPEP device **100**, it should be appreciated that the disclosed modifications may be applied to any of the embodiments described herein.

Fifth Embodiment

[0159] Referring to FIGS. **42-43**, an OPEP device **500** is shown having a chamber inlet bypass **578** adapted to permit exhaled air into a first chamber **514** without passing through the chamber inlet **504**. With the exception of the chamber inlet bypass **578**, the OPEP device **500** is otherwise configured and operates the same as the OPEP device **100**. As shown, the OPEP device **500** includes a housing **502** comprising a front section **501**, a middle section **503**, and a rear section **505**. The housing is also associated with a mouthpiece **509**, and includes a first chamber outlet **506** and a second chamber outlet (not shown) opposite the first chamber outlet **506**. As seen in the cross-sectional view of the FIG. **43**, the OPEP device **500** includes a restrictor member **530** positioned relative to a seat **524** about the chamber inlet **504** such that it is moveable between a closed position, where the flow of exhaled air through the chamber inlet **504** is restricted, and an open position, where the flow of exhaled air through the chamber inlet **504** is less restricted. The OPEP device **500** further includes a chamber inlet bypass **578** that allows a small amount of exhaled air to move past the chamber inlet **504** and the restrictor member **530** at all times. An exemplary flow path **577** through the chamber inlet bypass **578** is identified in FIG. **43** by a dashed line. By permitting a small amount of exhaled air to bypass the chamber inlet **504** and the restrictor member **530** through the chamber inlet bypass **578**, the amplitude of the OPEP therapy administered by the OPEP device **500** is decreased, while the frequency remains substantially unaffected.

[0160] Furthermore, a regulation member **579** extending from the mouthpiece **509** permits a user to

selectively adjust the amount of exhaled air allowed to flow through the chamber inlet bypass **578**. For example, as shown in FIG. **43**, a user may rotate the mouthpiece **509** relative to the front section **501** of the housing **502**, thereby rotating the regulation member **579** relative to the chamber inlet bypass **578**, to either increase or decrease the cross-sectional area of the chamber inlet bypass **578** through which exhaled air may flow. In this way, the user may selectively adjust the OPEP device **500** to maintain the ideal operating conditions.

Sixth Embodiment

[0161] Referring to FIGS. **44-45**, an OPEP device **600** is shown having a control port **680** adapted to permit exhaled air to exit the respiratory treatment device **600** prior to entering a first chamber **614** of the OPEP device **600**. With the exception of the control port **680**, the OPEP device **600** is otherwise configured and operates the same as the OPEP device **100**. As shown, the OPEP device **600** includes a housing **602** comprising a front section **601**, a middle section **603**, and a rear section **605**. The housing **602** is also associated with a mouthpiece **609**, and includes a first chamber outlet **606** and a second chamber outlet (not shown) positioned opposite the first chamber outlet **606**. As seen in the cross-sectional view of the FIG. **45**, the OPEP device **600** includes a restrictor member **630** positioned relative to a chamber inlet **604** such that it is moveable between a closed position, where the flow of exhaled air through the chamber inlet **604** is restricted, and an open position, where the flow of exhaled air through the chamber inlet **604** is less restricted, as well as a variable nozzle **636**, and a vane **632** operatively connected to the restrictor member **630** by a shaft (not shown). A control port **680** allows a small amount of exhaled air to exit the respiratory treatment device **600** prior to entering the first chamber **614** of the OPEP device **600**. An exemplary flow path **681** through the control port **680** is identified in FIG. **45** by a dashed line. By permitting a small amount of exhaled air to exit the OPEP device **600** through the control port **680**, the amplitude and the frequency of the OPEP therapy administered by the OPEP device **600** is decreased.

[0162] Furthermore, the mouthpiece **609** is rotatable relative to the front section **601** of the housing **602** to permit a user to selectively adjust the amount of exhaled air allowed to exit the respiratory treatment device **600** through the control port **680**. For example, as shown in FIG. **45**, a user may rotate the mouthpiece **609** relative to the front section **601** to either increase or decrease the cross-sectional area of the control port **680** through which exhaled air may flow. In this way, the user may selectively adjust the OPEP device **600** to maintain the ideal operating conditions.

[0163] Operating conditions similar to those described below with reference to the OPEP device **800** may also be achievable for an OPEP device according to the OPEP device **600**.

Seventh Embodiment

[0164] Turning to FIGS. **46-47**, an OPEP device **700** is shown having a first control port **780** adapted to permit exhaled air to exit the respiratory treatment device **700** prior to entering a first chamber **714**, and a second control port **782** adapted to permit exhaled air to exit the respiratory treatment device **700** from the first chamber **714**. With the exception of the first control port **780** and the second control port **782**, the OPEP device **700** is otherwise configured and operates the same as the OPEP device **100**. As shown, the OPEP device **700** includes a housing **702** comprising a front section **701**, a middle section **703**, and a rear section **705**. The housing is also associated with a mouthpiece **709**, and includes a first chamber outlet **706** and a second chamber outlet (not shown) positioned opposite the first chamber outlet **706**. As seen in the cross-sectional view of the FIG. **47**, the OPEP device **700** includes a restrictor member **730** positioned relative to a chamber inlet **704** such that it is moveable between a closed position, where the flow of exhaled air through the chamber inlet **704** is restricted, and an open position, where the flow of exhaled air through the chamber inlet **704** is less restricted, as well as a variable nozzle **736**, and a vane **732** operatively connected to the restrictor member **730** by a shaft (not shown).

[0165] Furthermore, both the first control port **780** and the second control port **782** may be equipped with regulation members **779**, **783** configured to permit a user to selectively adjust the amount of exhaled air allowed to exit the respiratory treatment device **700** through either the first

control port **780** or the second control port **782**. For example, as shown in FIGS. **46-47**, the regulation members **779, 783** are formed as a ring configured to rotate relative to the housing **702** to either increase or decrease the cross-sectional area of the control port **780, 782** through which exhaled air may flow. By selectively increasing the cross-sectional area of the first control port **780** through which exhaled air may flow, a user may decrease the amplitude and frequency of the OPEP therapy administered by the OPEP device **700**, and vice-versa. By selectively increasing the cross-sectional area of the second control port **782**, a user may decrease the frequency of the OPEP therapy administered by the OPEP device **700**, and vice-versa. In this way, a user may selectively adjust the OPEP device **700** to maintain the ideal operating conditions.

Eighth Embodiment

[0166] Turning to FIGS. **48-50**, another embodiment of an OPEP device **800** is shown. The OPEP device **800** is similar to that of the OPEP device **200** in that is selectively adjustable. As best seen in FIGS. **48, 50, 53**, and **62**, the OPEP device **800**, like the OPEP device **200**, includes an adjustment mechanism **853** adapted to change the relative position of a chamber inlet **804** with respect to a housing **802** and a restrictor member **830**, which in turn changes the range of rotation of a vane **832** operatively connected thereto. As previously explained with regards to the OPEP device **200**, a user is therefore able to conveniently adjust both the frequency and the amplitude of the OPEP therapy administered by the OPEP device **800** without opening the housing **802** and disassembling the components of the OPEP device **800**. The administration of OPEP therapy using the OPEP device **800** is otherwise the same as described above with regards to the OPEP device **100**.

[0167] The OPEP device **800** comprises a housing **802** having a front section **801**, a rear section **805**, and an inner casing **803**. As with the previously described OPEP devices, the front section **801**, the rear section **805**, and the inner casing **803** are separable so that the components contained therein can be periodically accessed, cleaned, or reconfigured, as required to maintain the ideal operating conditions. For example, as shown in FIGS. **48-50**, the front section **801** and the rear section **805** of the housing **802** are removably connected via a snap fit engagement.

[0168] The components of the OPEP device **800** are further illustrated in the exploded view of FIG. **51**. In general, in addition to the front section **801**, the rear section **805**, and the inner casing **803**, the OPEP device **800** further comprises a mouthpiece **809**, an inhalation port **811**, a one-way valve **884** disposed therebetween, an adjustment mechanism **853**, a restrictor member **830**, a vane **832**, and a variable nozzle **836**.

[0169] As seen in FIGS. **52-53**, the inner casing **803** is configured to fit within the housing **802** between the front section **801** and the rear section **805**, and partially defines a first chamber **814** and a second chamber **818**. The inner casing **803** is shown in further detail in the perspective and cross sectional views shown in FIGS. **54-55**. A first chamber outlet **806** and a second chamber outlet **808** are formed within the inner casing **803**. One end **885** of the inner casing **803** is adapted to receive the variable nozzle **836** and maintain the variable nozzle **836** between the rear section **805** and the inner casing **803**. An upper bearing **826** and a lower bearing **828** for supporting the adjustment mechanism **853** is formed, at least in part, within the inner casing **803**. Like the flexible cylinder **271** and sealing edge **270** described above with regards to the OPEP device **200**, the inner casing **803** also includes a flexible cylinder **871** with a sealing edge **870** for engagement about a frame **856** of the adjustment mechanism **853**.

[0170] The vane **832** is shown in further detail in the perspective view shown in FIG. **56**. A shaft **834** extends from the vane **832** and is keyed to engage a corresponding keyed portion within a bore **865** of the restrictor member **830**. In this way, the shaft **834** operatively connects the vane **832** with the restrictor member **830** such that the vane **832** and the restrictor member **830** rotate in unison.

[0171] The restrictor member **830** is shown in further detail in the perspective views shown in FIGS. **57-58**. The restrictor member **830** includes a keyed bore **865** for receiving the shaft **834** extending from the vane **832**, and further includes a stop **822** that limits permissible rotation of the restrictor member **830** relative to a seat **824** of the adjustment member **853**. As shown in the front

view of FIG. 59, like the restrictor member 130, the restrictor member 830 further comprises an offset designed to facilitate movement of the restrictor member 830 between a closed position and an open position. More specifically, a greater surface area of the face 840 of the restrictor member 830 is positioned on one side of the bore 865 for receiving the shaft 834 than on the other side of the bore 865. As described above with regards to the restrictor member 130, this offset produces an opening torque about the shaft 834 during periods of exhalation.

[0172] The adjustment mechanism 853 is shown in further detail in the front and rear perspective views of FIGS. 60 and 61. In general, the adjustment mechanism includes a frame 856 adapted to engage the sealing edge 870 of the flexible cylinder 871 formed on the inner casing 803. A circular opening in the frame 856 forms a seat 824 shaped to accommodate the restrictor member 830. In this embodiment, the seat 824 also defines the chamber inlet 804. The adjustment mechanism 853 further includes an arm 854 configured to extend from the frame 856 to a position beyond the housing 802 in order to permit a user to selectively adjust the orientation of the adjustment mechanism 853, and therefore the chamber inlet 804, when the OPEP device 800 is fully assembled. The adjustment mechanism 853 also includes an upper bearing 885 and a lower bearing 886 for receiving the shaft 834.

[0173] An assembly of the vane 832, the adjustment mechanism 853, and the restrictor member 830 is shown in the perspective view of FIG. 62. As previously explained, the vane 832 and the restrictor member 830 are operatively connected by the shaft 834 such that rotation of the vane 832 results in rotation of the restrictor member 830, and vice versa. In contrast, the adjustment mechanism 853, and therefore the seat 824 defining the chamber inlet 804, is configured to rotate relative to the vane 832 and the restrictor member 830 about the shaft 834. In this way, a user is able to rotate the arm 854 to selectively adjust the orientation of the chamber inlet 804 relative to the restrictor member 830 and the housing 802. For example, a user may increase the frequency and amplitude of the OPEP therapy administered by the OPEP device 800 by rotating the arm 854, and therefore the frame 856, in a clockwise direction. Alternatively, a user may decrease the frequency and amplitude of the OPEP therapy administered by the OPEP device 800 by rotating the adjustment arm 854, and therefore the frame 256, in a counter-clockwise direction. Furthermore, as shown for example in FIGS. 48 and 50, indicia may be provided on the housing 802 to aid the user in the setting of the appropriate configuration of the OPEP device 800.

[0174] The variable nozzle 836 is shown in further detail in the front and rear perspective views of FIGS. 63 and 64. The variable nozzle 836 in the OPEP device 800 is similar to the variable nozzle 236 described above with regards to the OPEP device 200, except that the variable nozzle 836 also includes a base plate 887 configured to fit within one end 885 (see FIGS. 54-55) of the inner casing 803 and maintain the variable nozzle 836 between the rear section 805 and the inner casing 803. Like the variable nozzle 236, the variable nozzle 836 and base plate 887 may be made of silicone.

[0175] The one-way valve 884 is shown in further detail in the front perspective view of FIG. 65. In general, the one-way valve 884 comprises a post 888 adapted for mounting in the front section 801 of the housing 802, and a flap 889 adapted to bend or pivot relative to the post 888 in response to a force or a pressure on the flap 889. Those skilled in the art will appreciate that other one-way valves may be used in this and other embodiments described herein without departing from the teachings of the present disclosure. As seen in FIGS. 52-53, the one-way valve 884 may be positioned in the housing 802 between the mouthpiece 809 and the inhalation port 811.

[0176] As discussed above in relation to the OPEP device 100, the OPEP device 800 may be adapted for use with other or additional interfaces, such as an aerosol delivery device. In this regard, the OPEP device 800 is equipped with an inhalation port 811 (best seen in FIGS. 48-49 and 51-53) in fluid communication with the mouthpiece 809. As noted above, the inhalation port may include a separate one-way valve 884 (best seen in FIGS. 52-53 and 65) configured to permit a user of the OPEP device 800 both to inhale the surrounding air through the one-way valve 884 and to exhale through the chamber inlet 804, without withdrawing the mouthpiece 809 of the OPEP

device **800** between periods of inhalation and exhalation. In addition, the aforementioned commercially available aerosol delivery devices may be connected to the inhalation port **811** for the simultaneous administration of aerosol therapy (upon inhalation) and OPEP therapy (upon exhalation).

[0177] The OPEP device **800** and the components described above are further illustrated in the cross-sectional views shown in FIGS. **52-53**. For purposes of illustration, the cross-sectional view of FIG. **52** is shown without all the internal components of the OPEP device **800**.

[0178] The front section **801**, the rear section **805**, and the inner casing **803** are assembled to form a first chamber **814** and a second chamber **818**. As with the OPEP device **100**, an exhalation flow path **810**, identified by a dashed line, is defined between the mouthpiece **809** and at least one of the first chamber outlet **806** (best seen in FIGS. **52-53** and **55**) and the second chamber outlet **808** (best seen in FIG. **54**), both of which are formed within the inner casing **803**. As a result of the inhalation port **811** and the one-way valve **848**, the exhalation flow path **810** begins at the mouthpiece **809** and is directed toward the chamber inlet **804**, which in operation may or may not be blocked by the restrictor member **830**. After passing through the chamber inlet **804**, the exhalation flow path **810** enters the first chamber **814** and makes a 180° turn toward the variable nozzle **836**. After passing through an orifice **838** of the variable nozzle **836**, the exhalation flow path **810** enters the second chamber **818**. In the second chamber **818**, the exhalation flow path **810** may exit the second chamber **818**, and ultimately the housing **802**, through at least one of the first chamber outlet **806** or the second chamber outlet **808**. Those skilled in the art will appreciate that the exhalation flow path **810** identified by the dashed line is exemplary, and that air exhaled into the OPEP device **800** may flow in any number of directions or paths as it traverses from the mouthpiece **809** or chamber inlet **804** to the first chamber outlet **806** or the second chamber outlet **808**. As previously noted, the administration of OPEP therapy using the OPEP device **800** is otherwise the same as described above with regards to the OPEP device **100**.

[0179] Solely by way of example, the follow operating conditions, or performance characteristics, may be achieved by an OPEP device according to the OPEP device **800**, with the adjustment dial **854** set for increased frequency and amplitude:

TABLE-US-00001 Flow Rate (Ipm) 10 30 Frequency (Hz) 7 20 Upper Pressure (cm H₂O) 13 30
Lower Pressure (cm H₂O) 1.5 9 Amplitude (cm H₂O) 11.5 21

[0180] The frequency and amplitude may decrease, for example, by approximately 20% with the adjustment dial **854** set for decreased frequency and amplitude. Other frequency and amplitude targets may be achieved by varying the particular configuration or sizing of elements, for example, increasing the length of the vane **832** results in a slower frequency, whereas, decreasing the size of the orifice **838** results in a higher frequency. The above example is merely one possible set of operating conditions for an OPEP device according to the embodiment described above.

Ninth Embodiment

[0181] Turning to FIGS. **66-69**, another embodiment of a respiratory treatment device **900** is shown. Unlike the previously described OPEP devices, the respiratory treatment device **900** is configured to administer oscillating pressure therapy upon both exhalation and inhalation. Those skilled in the art will appreciate that the concepts described below with regards to the respiratory treatment device **900** may be applied to any of the previously described OPEP devices, such that oscillating pressure therapy may be administered upon both exhalation and inhalation. Likewise, the respiratory treatment device **900** may incorporate any of the concepts above regarding the previously described OPEP devices, including for example, a variable nozzle, an inhalation port adapted for use with an aerosol delivery device for the administration of aerosol therapy, an adjustment mechanism, a chamber inlet bypass, one or more control ports, etc.

[0182] As shown in FIGS. **66** and **67**, the respiratory treatment device **900** includes a housing **902** having a front section **901**, a middle section **903**, and a rear section **905**. As with the OPEP devices described above, the housing **902** is openable so that the contents of the housing **902** may be

accessed for cleaning and/or selective replacement of the components contained therein to maintain ideal operating conditions. The housing **902** further includes a first opening **912**, a second opening **913**, and a third opening **915**.

[0183] Although the first opening **912** is shown in in FIGS. **66** and **67** in association with a mouthpiece **909**, the first opening **912** may alternatively be associated with other user interfaces, for example, a gas mask or a breathing tube. The second opening **913** includes a one-way exhalation valve **990** configured to permit air exhaled into the housing **902** to exit the housing **902** upon exhalation at the first opening **912**. The third opening **915** includes a one-way inhalation valve **984** configured to permit air outside the housing **902** to enter the housing **902** upon inhalation at the first opening **912**. As shown in greater detail in FIG. **67**, the respiratory treatment device **900** further includes a manifold plate **993** having an exhalation passage **994** and an inhalation passage **995**. A one-way valve **991** is adapted to mount to within the manifold plate **993** adjacent to the exhalation passage **994** such that the one-way valve **991** opens in response to air exhaled into the first opening **912**, and closes in response to air inhaled through the first opening **912**. A separate one-way valve **992** is adapted to mount within the manifold plate **993** adjacent to the inhalation passage **995** such that the one-way valve **992** closes in response to air exhaled into the first opening **912**, and opens in response to air inhaled through the first opening **912**. The respiratory treatment device **900** also includes a restrictor member **930** and a vane **932** operatively connected by a shaft **934**, the assembly of which may operate in the same manner as described above with regards to the disclosed OPEP devices.

[0184] Referring now to FIGS. **68** and **69**, cross-sectional perspective views are shown taken along lines I and II, respectively, in FIG. **66**. The respiratory treatment device **900** administers oscillating pressure therapy upon both inhalation and exhalation in a manner similar to that shown and described above with regards to the OPEP devices. As described in further detail below, the OPEP device **900** includes a plurality of chambers (i.e., more than one). Air transmitted through the first opening **912** of the housing **902**, whether inhaled or exhaled, traverses a flow path that passes, at least in part, past a restrictor member **930** housed in a first chamber **914**, and through a second chamber **918** which houses a vane **932** operatively connected to the restrictor member **930**. In this regard, at least a portion of the flow path for both air exhaled into or inhaled from the first opening **912** is overlapping, and occurs in the same direction.

[0185] For example, an exemplary flow path **981** is identified in FIGS. **68** and **69** by a dashed line. Similar to the previously described OPEP devices, the restrictor member **930** is positioned in the first chamber **914** and is movable relative to a chamber inlet **904** between a closed position, where the flow of air through the chamber inlet **904** is restricted, and an open position, where the flow of air through the chamber **904** inlet is less restricted. After passing through the chamber inlet **904** and entering the first chamber **914**, the exemplary flow path **981** makes a 180-degree turn, or reverses longitudinal directions (i.e., the flow path **981** is folded upon itself), whereupon the exemplary flow path **981** passes through an orifice **938** and enters the second chamber **918**. As with the previously described OPEP devices, the vane **932** is positioned in the second chamber **918**, and is configured to reciprocate between a first position and a second position in response to an increased pressure adjacent the vane, which in turn causes the operatively connected restrictor member **930** to repeatedly move between the closed position and the open position. Depending on the position of the vane **932**, air flowing along the exemplary flow path **981** is directed to one of either a first chamber outlet **906** or a second chamber outlet **908**. Consequently, as inhaled or exhaled air traverses the exemplary flow path **981**, pressure at the chamber inlet **904** oscillates.

[0186] The oscillating pressure at the chamber inlet **904** is effectively transmitted back to a user of the respiratory treatment device **900**, i.e., at the first opening **912**, via a series of chambers. As seen in FIGS. **68** and **69**, the respiratory treatment device includes a first additional chamber **996**, a second additional chamber **997**, and a third additional chamber **998**, which are described in further detail below.

[0187] The mouthpiece **909** and the first additional chamber **996** are in communication via the first opening **912** in the housing **902**. The first additional chamber **996** and the second additional chamber **997** are separated by the manifold plate **993**, and are in communication via the exhalation passage **994**. The one-way valve **991** mounted adjacent to the exhalation passage **994** is configured to open in response to air exhaled into the first opening **912**, and close in response to air inhaled through the first opening **912**.

[0188] The first additional chamber **996** and the third additional chamber **998** are also separated by the manifold plate **993**, and are in communication via the inhalation passage **995**. The one-way valve **992** mounted adjacent to the inhalation passage **995** is configured to close in response to air exhaled into the first opening **912**, and open in response to air inhaled through the first opening **912**.

[0189] Air surrounding the respiratory treatment device **900** and the second additional chamber **997** are in communication via the third opening **915** in the housing **902**. The one-way valve **984** is configured to close in response to air exhaled in to the first opening **912**, and open in response to air inhaled through the first opening **912**.

[0190] Air surrounding the respiratory treatment device **900** and the third additional chamber **998** are in communication via the second opening **913** in the housing **902**. The one way-valve **990** mounted adjacent the second opening **913** is configured to open in response to air exhaled into the first opening **912**, and close in response to air inhaled through the first opening **912**. The third additional chamber **998** is also in communication with the second chamber **918** via the first chamber outlet **906** and the second chamber outlet **908**.

[0191] Referring now to FIGS. **70-71**, cross-sectional perspective views taken along lines I and II, respectively, of FIG. **66**, illustrate an exemplary exhalation flow path **910** formed between the first opening **912**, or the mouthpiece **909**, and the second opening **913**. In general, upon exhalation by a user into the first opening **912** of the housing **902**, pressure builds in the first additional chamber **996**, causing the one-way valve **991** to open, and the one-way valve **992** to close. Exhaled air then enters the second additional chamber **997** through the exhalation passage **994** and pressure builds in the second additional chamber **997**, causing the one-way valve **984** to close and the restrictor member **930** to open. The exhaled air then enters the first chamber **914** through the chamber inlet **904**, reverses longitudinal directions, and accelerates through the orifice **938** separating the first chamber **914** and the second chamber **918**. Depending on the orientation of the vane **932**, the exhaled air then exits the second chamber **918** through one of either the first chamber outlet **906** or the second chamber outlet **908**, whereupon it enters the third additional chamber **998**. As pressure builds in the third additional chamber **998**, the one-way valve **990** opens, permitting exhaled air to exit the housing **902** through the second opening **913**. Once the flow of exhaled air along the exhalation flow path **910** is established, the vane **932** reciprocates between a first position and a second position, which in turn causes the restrictor member **930** to move between the closed position and the open position, as described above with regards to the OPEP devices. In this way, the respiratory treatment device **900** provides oscillating therapy upon exhalation.

[0192] Referring now to FIGS. **72-73**, different cross-sectional perspective views taken along lines I and II, respectively, of FIG. **66**, illustrate an exemplary inhalation flow path **999** formed between the third opening **915** and the first opening **912**, or the mouthpiece **909**. In general, upon inhalation by a user through the first opening **912**, pressure drops in the first additional chamber **996**, causing the one-way valve **991** to close, and the one-way valve **992** to open. As air is inhaled from the third additional chamber **998** into the first additional chamber **996** through the inhalation passage **995**, pressure in the third additional chamber **998** begins to drop, causing the one-way valve **990** to close. As pressure continues to drop in the third additional chamber **998**, air is drawn from the second chamber **918** through the first chamber outlet **906** and the second chamber outlet **908**. As air is drawn from the second chamber **918**, air is also drawn from the first chamber **914** through the orifice **938** connecting the second chamber **918** and the first chamber **914**. As air is drawn from the

first chamber **914**, air is also drawn from the second additional chamber **997** through the chamber inlet **904**, causing the pressure in the second additional chamber **997** to drop and the one-way valve **984** to open, thereby permitting air to enter the housing **902** through third opening **915**. Due to the pressure differential between the first additional chamber **996** and the second additional chamber **997**, the one-way valve **991** remains closed. Once the flow of inhaled air along the inhalation flow path **999** is established, the vane **932** reciprocates between a first position and a second position, which in turn causes the restrictor member **930** to move between the closed position and the open position, as described above with regards to the OPEP devices. In this way, the respiratory treatment device **900** provides oscillating therapy upon inhalation.

[0193] Referring now to FIG. **74**, a front perspective view is shown of the OPEP device **800** connected with an aerosol delivery device in the form of a nebulizer **899** via the inhalation port **811**. The system comprising the OPEP device **800** connected to the nebulizer **899** is configured to provide both oscillating pressure therapy and aerosol therapy, as described above. The combination of the OPEP device **800** and the nebulizer **899**, however, is exemplary. Alternative combinations of the OPEP devices described herein and aerosol delivery devices, such as those identified above, are also envisioned.

[0194] Those skilled in the art will appreciate that the various concepts described above with regards to a particular embodiment of an OPEP device may also be applied to any of the other embodiments described herein, even though not specifically shown or described with regards to the other embodiments. For example, any one of the embodiments described herein may include a variable nozzle, an inhalation port adapted for use with an aerosol delivery device for the administration of aerosol therapy, an adjustment mechanism for adjusting the relative position of the chamber inlet and/or the permissible range of movement by a restrictor member, a chamber inlet bypass, one or more control ports, etc.

[0195] Although the foregoing description is provided in the context of an OPEP device, it will also be apparent to those skilled in the art that any respiratory device may benefit from various teachings contained herein. The foregoing description has been presented for purposes of illustration and description, and is not intended to be exhaustive or to limit the inventions to the precise forms disclosed. It will be apparent to those skilled in the art that the present inventions are susceptible of many variations and modifications coming within the scope of the following claims.

EXEMPLARY IMPLEMENTATIONS

[0196] In one implementation, a respiratory treatment device includes a housing enclosing at least one chamber, a chamber inlet configured to receive exhaled air into the at least one chamber, and at least one chamber outlet configured to permit exhaled air to exit the at least one chamber. An exhalation flow path is defined between the chamber inlet and the at least one chamber outlet, and a restrictor member is positioned in the exhalation flow path, the restrictor member being moveable between a closed position, where a flow of exhaled air through the chamber inlet is restricted, and an open position, where the flow of exhaled air through the chamber inlet is less restricted. An orifice is positioned along the exhalation flow path through which the exhaled air passes, and a vane is positioned adjacent the orifice, the vane being operatively connected to the restrictor member, and configured to reciprocate between a first position and a second position in response to an increased pressure adjacent the vane. Additionally, the restrictor member moves between the closed position and the open position in response to the vane reciprocating between the first position and the second position.

[0197] The restrictor member may be positioned in a first chamber, and the vane may be positioned in a second chamber. The orifice may connect the first chamber and the second chamber. A size of the orifice may be configured to change in response to the flow of exhaled air through the orifice. The restrictor member may be a butterfly valve. The vane may be operatively connected to the restrictor member by a shaft. A face of the restrictor member may be rotatable about an axis of rotation, and the face of the restrictor member may be radially offset from the axis of rotation. The

face of the restrictor member may also have a greater surface area positioned on one side of the shaft than on the other side of the shaft. An orientation of the chamber inlet may be selectively adjustable. A chamber inlet bypass may be configured to permit exhaled air into the at least one chamber without passing through the chamber inlet. A control port may be configured to permit exhaled air to exit the respiratory treatment device prior to entering the at least one chamber. A control port may also be configured to permit exhaled air to exit the first chamber. An inhalation port may be in fluid communication with a user interface, and a one-way valve may be configured to permit air to flow through the inhalation port to the user interface upon inhalation. The inhalation port may also be configured to receive an aerosol medicament from an aerosol delivery device. The exhalation flow path may be folded upon itself.

[0198] In another implementation, a method of performing OPEP therapy includes receiving a flow of exhaled air along an exhalation flow path defined between an inlet and an outlet of a respiratory treatment device, directing the flow of exhaled air toward a vane, reciprocating the vane between a first position and a second position in response to the flow of exhaled air, and, moving a restrictor member in response to the reciprocal movement of the vane between a closed position, where a flow of exhaled air through the chamber inlet is restricted, and an open position, where the flow of exhaled air through the chamber inlet is less restricted.

[0199] In another implementation, a method of performing OPEP therapy includes receiving a flow of exhaled air along an exhalation flow path defined between an inlet and an outlet of a respiratory treatment device, accelerating the flow of exhaled air through an orifice positioned along the exhalation flow path, reciprocating a vane adjacent the orifice between a first position and a second position in response to the flow of exhaled air through the orifice, and, moving a restrictor member in response to the reciprocal movement of the vane between a closed position, where the flow of exhaled air along the exhalation flow path is restricted, and an open position, where the flow of exhaled air along the exhalation flow path is less restricted. The method may also include changing a size of the orifice in response to the flow of exhaled air through the orifice.

Claims

1-19. (canceled)

20. A respiratory treatment device comprising: a housing; an inlet configured to receive air into the housing; an outlet configured to permit air to exit the housing; a flow path defined between the inlet and the outlet; an opening positioned along the flow path, the flow path passing through the opening; and, a valve movable relative to the opening between a closed position, where the flow of air through the opening is restricted, and an open position, where the flow of air through the opening is less restricted; wherein a position of the opening is selectively adjustable.

21. The respiratory treatment device of claim 20, wherein the position of the opening is selectively adjustable to adjust a range of motion of the valve.

22. The respiratory treatment device of claim 20, wherein the opening is rotatable.

23. The respiratory treatment device of claim 20, wherein the valve is rotatable relative to the opening between the closed position, where the flow of air through the opening is restricted, and the open position, where the flow of air through the opening is less restricted.

24. The respiratory treatment device of claim 23, wherein the position of the opening is selectively adjustable to adjust a range of rotation of the valve.

25. The respiratory treatment device of claim 23, wherein the valve is prevented from completing a full revolution.

26. The respiratory treatment device of claim 20, wherein the valve is a butterfly valve.

27. The respiratory treatment device of claim 20, further comprising a vane in communication with the flow path, the vane operatively connected to and configured to move the valve in response to the flow of air along the flow path.

- 28.** The respiratory treatment device of claim 20, wherein selective adjustment of the position of the opening adjusts a frequency of oscillating pressure at the inlet during a flow of air along the flow path.
- 29.** The respiratory treatment device of claim 20, wherein selective adjustment of the position of the opening adjusts an amplitude of oscillating pressure at the inlet a flow of air along the flow path.
- 30.** A respiratory treatment device comprising: a housing; an inlet configured to receive air into the housing; an outlet configured to permit air to exit the housing; a flow path defined between the inlet and the outlet; an opening positioned along the flow path, the flow path passing through the opening; and, a valve movable relative to the opening between a closed position, where the flow of air through the opening is restricted, and an open position, where the flow of air through the opening is less restricted; wherein a range of motion of the valve is selectively adjustable.
- 31.** The respiratory treatment device of claim 30, wherein a position of the opening is selectively adjustable to adjust the range of motion of the valve.
- 32.** The respiratory treatment device of claim 31, wherein the opening is rotatable.
- 33.** The respiratory treatment device of claim 30, wherein the valve is rotatable relative to the opening between the closed position, where the flow of air through the opening is restricted, and the open position, where the flow of air through the opening is less restricted.
- 34.** The respiratory treatment device of claim 33, wherein a position of the opening is selectively adjustable to adjust a range of rotation of the valve.
- 35.** The respiratory treatment device of claim 30, wherein the valve is prevented from completing a full revolution.
- 36.** The respiratory treatment device of claim 30, wherein the valve is a butterfly valve.
- 37.** The respiratory treatment device of claim 30, further comprising a vane in communication with the flow path, the vane operatively connected to and configured to move the valve in response to the flow of air along the flow path.
- 38.** The respiratory treatment device of claim 30, wherein selective adjustment of the range of motion of the valve adjusts a frequency of oscillating pressure at the inlet a flow of air along the flow path.
- 39.** The respiratory treatment device of claim 30, wherein selective adjustment of the range of motion of the valve adjusts an amplitude of oscillating pressure at the inlet a flow of air along the flow path.
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